

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 9 – 20 March 2026 ¹

1. Additional example

- (a) Apply Itô's lemma to W_t^4 and calculate $\mathbb{E}(W_t^4)$. Show that for a normally distributed random variable $X \sim \mathcal{N}(0, \sigma^2)$ its fourth moment is $\mathbb{E}(X^4) = 3\sigma^4$.
- (b) We have seen that

$$I_t(W_t) = \int_0^t w_u dW_u = \frac{1}{2}(W_t^2 - t).$$

Determine $\mathbb{E}(I_t(W_t))$ and $\text{Var}(I_t(W_t))$ using Lemma 4.12, then check that the result is correct by calculating expected value and variance of the RHS.

- (c) Apply Itô's lemma to tW_t^2 and use this to show that

$$\int_0^t W_u^2 du = tW_t^2 - \int_0^t u du - \int_0^t 2uW_u dW_u.$$

2. (2020 exam, Q4 (a) to (c)) Let X be an Itô process of the form $dX_t = b_t dt + \sigma_t dW_t$. You may assume throughout this question that Itô's lemma holds in the form:

$$df(t, X_t) = \left[f_t(t, X_t) + b_t f_x(t, X_t) + \frac{1}{2} \sigma_t^2 f_{xx}(t, X_t) \right] dt + \sigma_t f_x(t, X_t) dW_t.$$

- (a) Derive a simplified form of Itô's lemma in the case where $X_t = W_t$ is just standard Brownian motion. Give a necessary and sufficient condition in terms of f_t , f_x and f_{xx} for the process $f(t, W_t)$ to be a martingale.
- (b) **(5 marks)**

Use Itô's lemma to put the following stochastic processes in standard form $dX_t = b_t dt + \sigma_t dW_t$:

- i. $W_t^3 - t^2$,
- ii. $\exp(3W_t)$,
- iii. $\exp(\sigma W_t + at)$.

Deduce a necessary and sufficient condition on a for process iii. to be a martingale.

- (c) Give a complete list of the functions of Brownian motion alone, $g(W_t)$, which are martingales.

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